

Convex optimization

1 – Introduction

Optimization? Convexity?

1.1 Optimization

Definition

- From Wikipedia: *"the selection of a best element (with regard to some criterion) from some set of available alternatives"*.
- Examples:
 - › Find the fastest route from one metro station to another.
 - › Determine the optimal investment among several stocks to maximize the potential gain while limiting risks.
 - › Choose which products to buy to best meet a need (e.g. lifespan, quality/price ratio) while meeting constraints (e.g. budget).

Definition: a more practical one

- For our purposes: maximizing or minimizing a real function f over a set A .

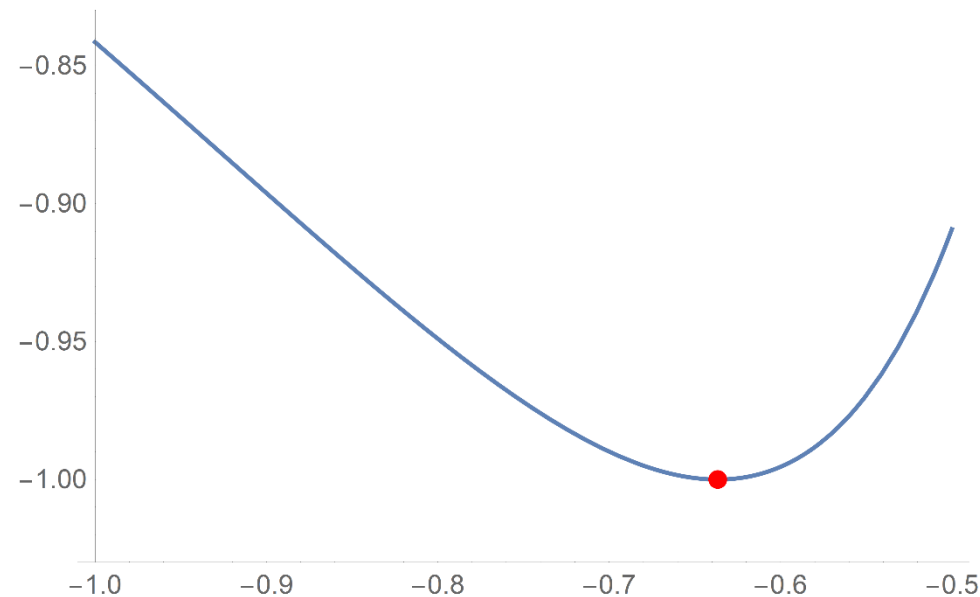
$$f: A \rightarrow \mathbb{R}$$

$$\underset{\mathbf{x} \in A}{\text{minimize}} \quad f(\mathbf{x})$$

$$\underset{\mathbf{x} \in A}{\text{maximize}} \quad f(\mathbf{x})$$

Definition: a more practical one

- For our purposes: maximizing or minimizing a real function f over a set A .



Applications

- Economics and finance
 - › maximize profit, minimize loss
- Engineering
 - › maximize efficiency, minimize error
- Operations research
 - › maximize performance, minimize risk
- Machine learning
 - › minimize the discrepancy between a model (explanatory or predictive) and reality (ground truth data)
- And so on...

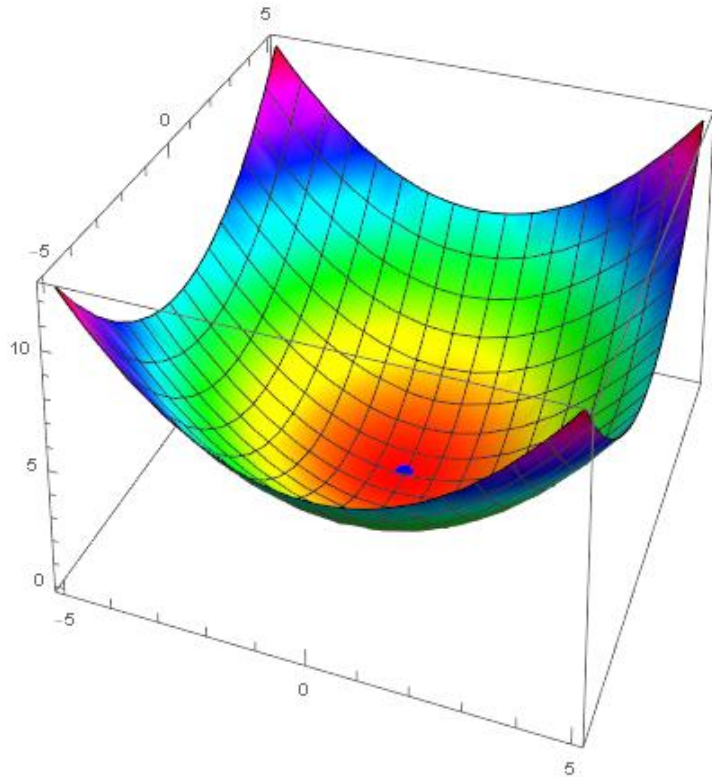
Notation

$$\underset{\mathbf{x} \in A}{\text{minimize}} \quad f(\mathbf{x})$$

$$\underset{\mathbf{x} \in A}{\text{maximize}} \quad f(\mathbf{x})$$

- The function $f: A \rightarrow \mathbb{R}$ is called the **objective function** (cost function if it is a minimization).
- The set A is called the **feasible set**. It is, typically, a subset of the vector space $\mathbb{R}^n$.
- An element $\mathbf{x} \in A$ is called a **feasible solution**.
- A feasible solution that minimizes or maximizes (according to the objective pursued) the objective function is called an **optimal solution**, and denoted $\mathbf{x}^*$.

Example: minimization



- $f(\mathbf{x}) = f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$
- $A = \mathbb{R}^2$
- $\mathbf{x}^* = (x^*, y^*) = (0, 0)$
- $f(\mathbf{x}^*) = 0$

Notation

- Usually, the feasible set is described as the vector space $\mathbb{R}^n$ with some inequality constraints.
- One can then write an optimization problem as:

$$\begin{array}{ll}\underset{\mathbf{x} \in \mathbb{R}^n}{\text{minimize}} & f(\mathbf{x}) \\ \text{subject to} & c_i(\mathbf{x}) \leq b_i, \quad i = 1, \dots, m\end{array}$$

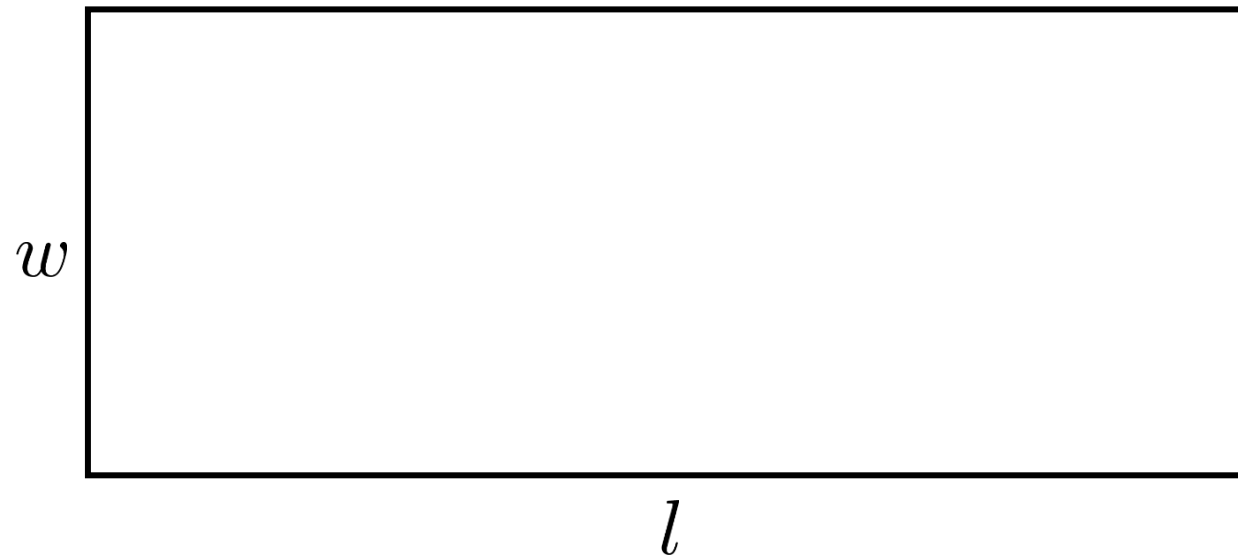
where $\mathbf{x} = (x_1, \dots, x_n)$ is the vector of **optimization variables**, the functions $c_1, \dots, c_m: \mathbb{R}^n \rightarrow \mathbb{R}$ are the **constraints functions**, and the constants $b_1, \dots, b_m$ are the **bounds**.

Modeling

- An important aspect of optimization is to translate real life problems into formalized ones
- What are the quantities whose values we are trying to obtain?
 - › What are the optimization variables?
- What is the value we want to optimize (minimize or maximize)?
 - › What is the objective function?
- What are the constraints on the quantities sought?
 - › What is the feasible set?

Modeling example: optimal rectangle

- We are looking to build a rectangle with the largest possible area and whose perimeter is at most 42 cm.



Modeling example: optimal rectangle

- What are the quantities whose values we are trying to obtain?
 - › w and l : the optimization variables
- What is the value we want to optimize (minimize or maximize)?
 - › Maximize the area $w \times l$: the objective function
- What are the constraints on the quantities sought?
 - › The perimeter $2w + 2l$ must be less than or equal to 42, and both variables must be positive : the feasible set

Modeling example: optimal rectangle

- We are looking to build a rectangle with the largest area possible and whose perimeter is at most 42 cm.

$$\begin{array}{ll}\text{maximize} & w \times l \\ (w,l) \in \mathbb{R}^2 & \\ \text{subject to} & 2w + 2l \leq 42 \\ & w > 0, l > 0\end{array}$$

Modeling example: transportation problem

- An aircraft manufacturer wants to optimize (minimize the cost of) the travel of aluminum units from its 3 warehouses to its 2 fuselage manufacturing sites.
- Fuselage manufacturing sites: F_A, F_B
- Warehouses: W_1, W_2, W_3

	F_A	F_B
W_1	5	26
W_2	30	2
W_3	31	22

Cost of moving a Al unit from a warehouse to a fuselage manufacturing site

W_1	1291
W_2	2422
W_3	679

Al units available in warehouses

F_A	2403
F_B	1986

Al units required to operate a fuselage manufacturing site

Modeling example: transportation problem

- *What are the quantities whose values we are trying to obtain?*

The quantity of aluminum units to be moved from a given warehouse to a given manufacturing site. There are 6 possibilities:

$$W_1 \rightarrow F_A (x_{1,A})$$

$$W_1 \rightarrow F_B (x_{1,B})$$

$$W_2 \rightarrow F_A (x_{2,A})$$

$$W_2 \rightarrow F_B (x_{2,B})$$

$$W_3 \rightarrow F_A (x_{3,A})$$

$$W_3 \rightarrow F_B (x_{3,B})$$

Hence, we have 6 optimizations variables:

$$x_{1,A} \quad x_{2,A} \quad x_{3,A} \quad x_{1,B} \quad x_{2,B} \quad x_{3,B}$$

Modeling example: transportation problem

- *What is the value we want to optimize (minimize or maximize)?*

Minimize the cost of transportation c .

	F_A	F_B
W_1	5	26
W_2	30	2
W_3	31	22

Cost of moving a AI unit from a
warehouse to a fuselage
manufacturing site

$$c = 5x_{1,A} + 26x_{1,B} + 30x_{2,A} + 2x_{2,B} + 31x_{3,A} + 22x_{3,B}$$

Modeling example: transportation problem

- *What are the constraints on the quantities sought?*

W_1	1291
W_2	2422
W_3	679

AI units available in
warehouses

$$x_{1,A} + x_{1,B} \leq 1291$$

$$x_{2,A} + x_{2,B} \leq 2422$$

$$x_{3,A} + x_{3,B} \leq 679$$

F_A	2403
F_B	1986

AI units required to
operate a fuselage
manufacturing site

$$x_{1,A} + x_{2,A} + x_{3,A} \geq 2403$$

$$x_{1,B} + x_{2,B} + x_{3,B} \geq 1986$$

In addition, all quantities must be positive.

Modeling example: transportation problem

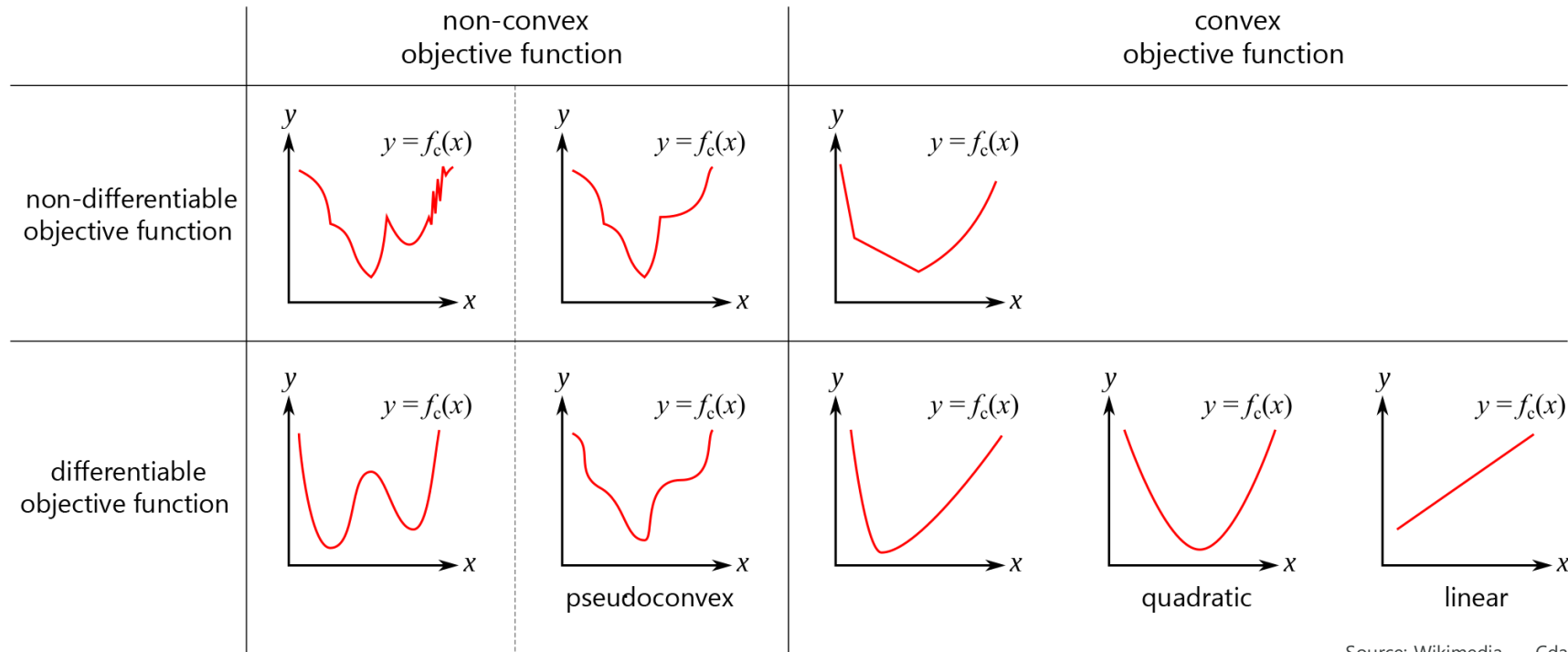
Hence, we obtain the following optimization problem:

$$\begin{aligned} & \underset{(x_{1,A}, \dots, x_{3,B}) \in \mathbb{R}^6}{\text{minimize}} && 5x_{1,A} + 26x_{1,B} + 30x_{2,A} + 2x_{2,B} + 31x_{3,A} + 22x_{3,B} \\ & \text{subject to} && x_{1,A} + x_{1,B} \leq 1291 \\ & && x_{2,A} + x_{2,B} \leq 2422 \\ & && x_{3,A} + x_{3,B} \leq 679 \\ & && x_{1,A} + x_{2,A} + x_{3,A} \geq 2403 \\ & && x_{1,B} + x_{2,B} + x_{3,B} \geq 1986 \\ & && x_{i,j} \geq 0, \quad \forall i \in \{1, 2, 3\}, \forall j \in \{A, B\} \end{aligned}$$

1.2 Convexity

Different types of optimization problems

- Different optimization problems can be grouped together according to the shape of their **objective function** and their **feasible set**.



Convexity

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Convexity, intuitively

- Convex optimization problems are in a way the broadest class of problems that can be solved quite efficiently.
- No risk of falling into a sub-optimal “hollow” (“easy to optimize”).

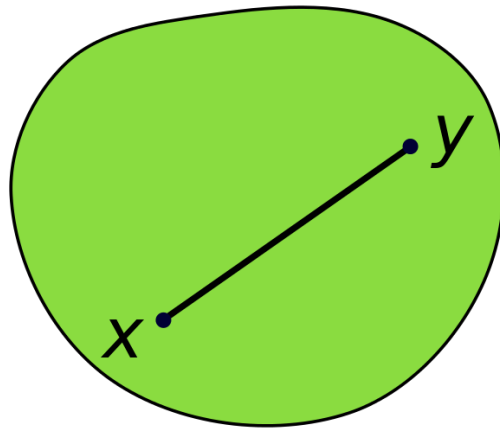
Convex optimization problem

$$\underset{\mathbf{x} \in A}{\text{minimize}} \quad f(\mathbf{x})$$

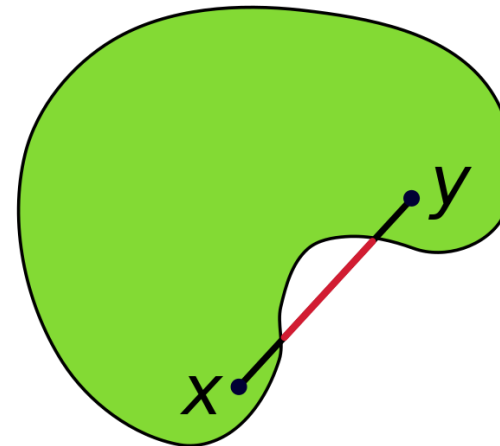
- A convex optimization problem is an optimization problem in which:
 - › The objective function f is a **convex function**.
 - › The feasible set A is a **convex set**.

Convex set

- Intuitively, a set is convex if it is closed and contains no hollows or bumps.
- More formally, for every pair of points within the set, every point on the straight line segment that joins the pair of points is also within the set.



Convex set

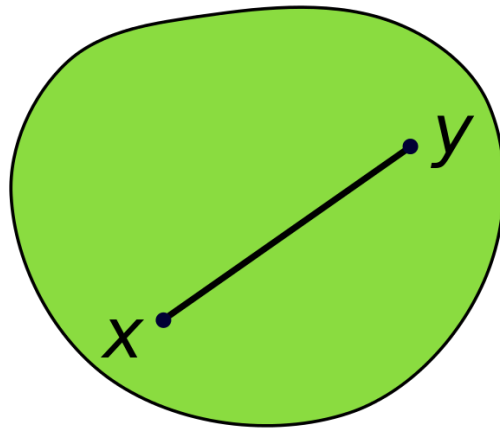


Non-convex set

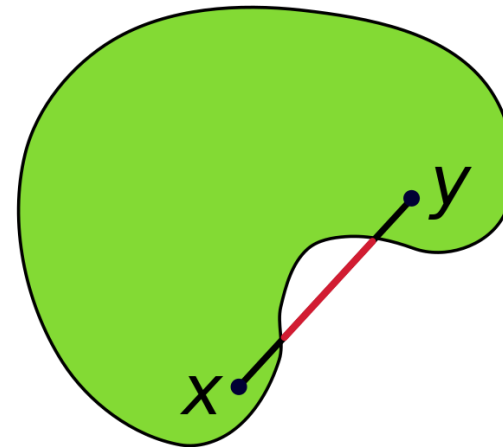
Convex set

- A set $A \subseteq \mathbb{R}^n$ is convex iff for every pair of points within the set, every point on the straight line segment that joins the pair of points is also within the set:

$$\forall x, y \in A, \forall \lambda \in [0, 1] : \lambda x + (1 - \lambda) y \in A$$



Convex set



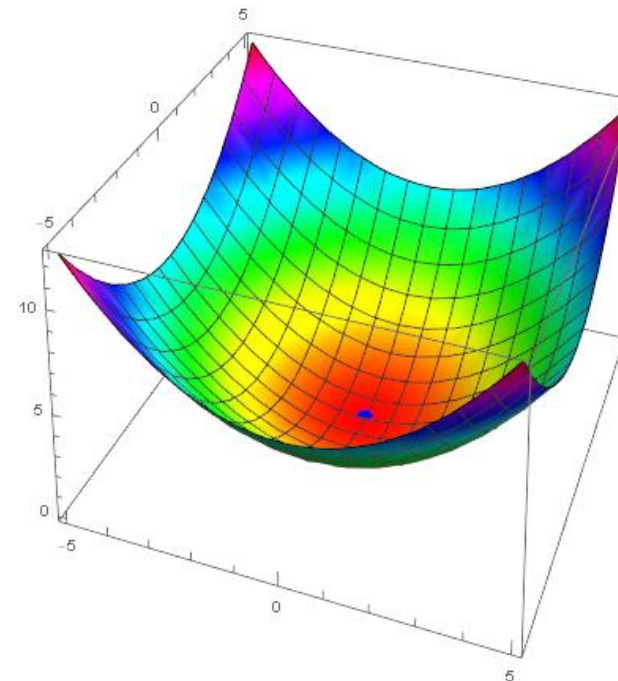
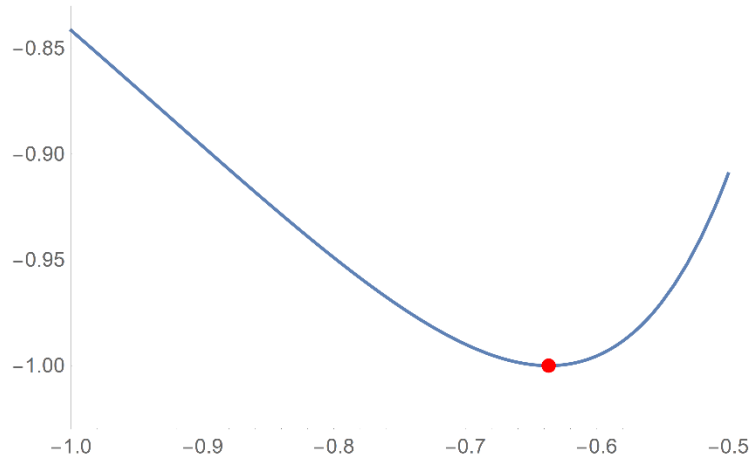
Non-convex set

Some properties about convex sets...

- A half-space is a convex set.
- The intersection of convex sets is a convex set.

Convex function (geometric interpretation)

- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex iff for every pair of points of the function graph, the straight line segment that joins the pair of points **is entirely located above or on** the graph.



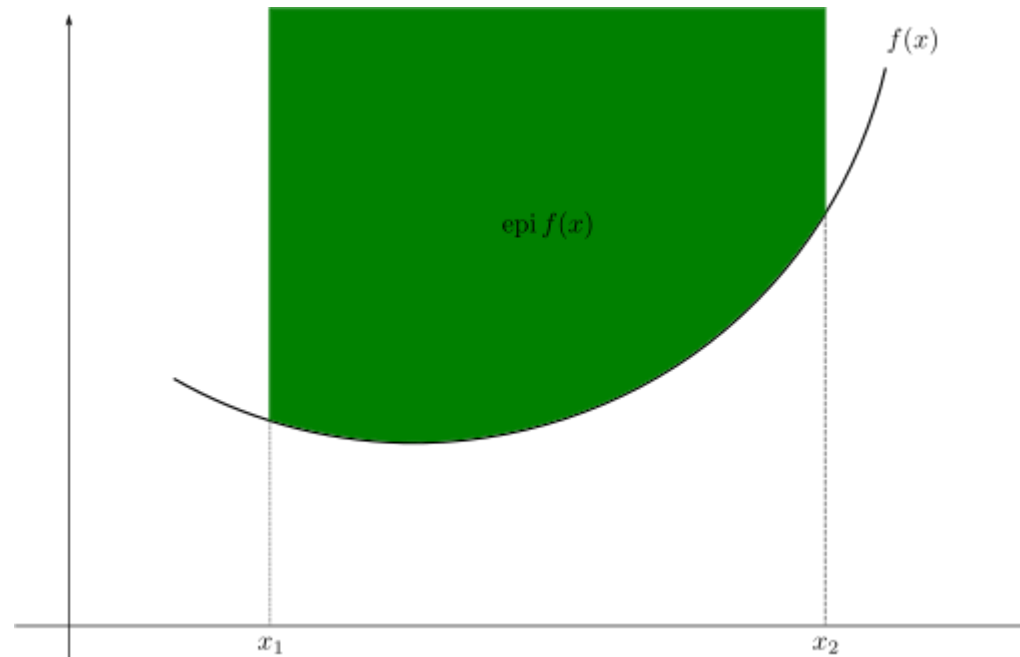
Convex function (formula)

- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex iff:

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \forall \lambda \in [0, 1] : f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y})$$

Convex function (epigraph)

- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex iff the set of points lying on or above its graph (called the **epigraph**) is a convex set.



Convex function (differential)

- A **twice differentiable** function of **one variable** $f: \mathbb{R} \rightarrow \mathbb{R}$ is convex iff:

$$\forall x \in \mathbb{R} : f''(x) \geq 0$$

Convex function (midpoint convexity)

- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is midpoint convex iff:

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n : f\left(\frac{\mathbf{x} + \mathbf{y}}{2}\right) \leq \frac{f(\mathbf{x})}{2} + \frac{f(\mathbf{y})}{2}$$

- Theorem: **a continuous midpoint convex function *is convex*.**

Gold property of convex functions

- **A local minimum of a convex function is also a global minimum.**

Concave functions

- The **concavity** is the opposite of convexity.
- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is concave iff $-f$ is convex.
- Maximizing a concave function is equivalent to minimizing a convex one.